Query-Performance Signals Do Not Route Between Sparse, Dense and Hybrid Retrieval: A Pre-Declared Negative Result on BEIR

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***Abstract*—Should the choice between lexical (BM25), dense and hybrid retrieval be made per domain or per query? We test both options on eight BEIR collections, using paired randomization tests, Holm correction and pre-declared decision rules. The domain-level hypothesis that motivated this project, namely that BM25 degrades on jargon-heavy collections, finds no support ($\rho = +0.33$, $p = 0.42$, $n = 8$), although eight domains can reveal only large effects. A per-query oracle over a fixed BM25+BGE-small hybrid would gain $+0.064$ nDCG@10, 3.4 times the collection-level headroom; this upper bound is inflated by per-query noise. A router trained leave-one-dataset-out on standard query-performance-prediction (QPP) features does not beat the always-hybrid baseline: macro nDCG@10 is 0.525 vs. 0.534, and on the six collections with more than 50 queries the difference is $+0.004$ (95% CI [-0.001, $+0.008$]). On five collections, three further methods also fail to beat it: a per-query fusion weight, a label-free weight tuned by a cross-encoder teacher, and cross-encoder re-ranking of the hybrid (on $\leq$150 queries per collection). Trained in-domain, the router recovers only which fixed system is best for each collection. The QPP features track query difficulty shared by all retrievers, not which retriever suits a query. In an exploratory analysis, choosing one system per collection from 10–20 labelled queries gains $+0.010$ to $+0.014$, more than any label-free router; almost all of this gain comes from Touché-2020, whose judgement pool favours BM25.**

***Index Terms*—BM25, dense retrieval, hybrid retrieval, query performance prediction, retriever selection, BEIR, negative results**

I. Introduction

Lexical retrievers such as BM25 [1] suffer from vocabulary mismatch [2]. Dense bi-encoders avoid it in principle, yet under zero-shot domain shift BM25 remains a strong baseline [3], and lexical–dense hybrids are often the most robust choice [4]–[6]. A practitioner facing a new collection must still decide which retriever to deploy, and whether to decide once per collection or per query.

This project began with a domain-level hypothesis: BM25 should degrade, relative to dense retrieval, on collections with specialised vocabulary (a high share of tokens that are rare in general English). The data do not support it. Across eight BEIR domains, this "jargon rate" was not associated with BM25 effectiveness in the expected direction ($\rho = +0.33$, $p = 0.42$). If anything, BM25 was *relatively* stronger on jargon-heavy collections ($\rho = -0.60$, $p = 0.12$, for the dense gain). Within each collection, by contrast, the lexical coverage of a query's relevant documents predicted BM25 well. This suggested that the query, not the domain, decides, and that a router could send each query to BM25, a dense model or a hybrid. We test that idea under pre-declared decision rules. Our contributions are:

- **A pre-declared negative result on query-time routing.** QPP-based routers trained leave-one-dataset-out (LODO) do not beat a fixed, untuned hybrid in any of seven configurations (four feature/model settings, three with percentile features). We report bootstrap bounds and a small-collection sensitivity analysis. Three further methods, on five collections, also fail to beat it: a continuous fusion weight, a label-free cross-encoder-tuned weight, and cross-encoder re-ranking.
- **An in-domain ceiling analysis.** Even with in-domain labels, the same features recover only the *collection-level* choice of system, except on the one collection with 8k training queries. We also calibrate the per-query oracle against per-query noise.
- **A difficulty-vs-lexical decomposition of the overlap–effectiveness link.** About two thirds of it is query difficulty shared with dense retrievers. The lexical-specific part needs relevance labels, and its label-free proxy carries no signal.
- **Effect sizes for robust defaults.** We reproduce Pyserini's flat BM25 and report effect sizes for lexical fixes, dense models and hybrids.

II. Related Work

BEIR and reproducibility. BEIR [3] evaluates retrievers zero-shot on 18 datasets, using multifield Anserini BM25 ($k_1 = 0.9$, $b = 0.4$). BM25 is a strong baseline there. BM25 followed by a cross-encoder beats BM25 on 16 of the 18 datasets, failing on ArguAna and Touché-2020. Kamalloo et al. [7] provide reproducible Pyserini reference runs, including the "flat" (title+text) BM25 that we reproduce. An earlier version of their paper names two open problems: significance testing across a heterogeneous benchmark, and shallow, possibly lexically biased judgement pools. On Touché-2020, neural models retrieve many short, unjudged passages; after denoising, neural nDCG@10 improves by up to 0.52, but BM25 remains more effective [8].

TABLE I
COLLECTIONS. $|q|$: MEAN QUERY LENGTH (TOKENS); REL/Q: RELEVANT DOCUMENTS PER QUERY; JARGON: CORPUS JARGON RATE.

Dataset	Domain	Docs	Queries	$\|q\|$	rel/q	jargon
NFCorpus	nutrition/med.	3,633	323	3.4	38.2	0.121
SciFact	scientific claims	5,183	300	13.1	1.1	0.186
FiQA	finance QA	57,638	648	11.2	2.6	0.034
SciDocs	citations	25,657	1,000	10.0	4.9	0.085
ArguAna	counter-args	8,674	1,406	195.8	1.0	0.033
TREC-COVID	COVID papers	171,332	50	11.7	493.5	0.151
Quora	duplicate qs	522,931	10,000	9.6	1.6	0.051
Touché-2020	arguments	382,545	49	6.7	19.0	0.042

Vocabulary mismatch and QPP. Term-mismatch diagnosis [9] and difficulty-based selective expansion [10] address the vocabulary problem in lexical retrieval. Formal et al. [11] find that neural IR fails to generalise term importance out of domain. QPP estimates effectiveness without labels, before retrieval [12], [13] or from the result list (Clarity [14], WIG [15], NQC [16], SMV [17]). We use the definitions collected by Meng et al. [18]. Faggioli et al. [19] report that QPP is less accurate for neural systems and, in lexical-oriented scenarios, fails on the queries where lexical and neural systems disagree most.

Selecting and fusing retrievers. Arabzadeh et al. [20] fine-tune BERT classifiers to choose, per query, between sparse and dense retrieval, or between sparse retrieval and a pooled hybrid. The classifiers are trained and tested in-domain on MS MARCO with recall@1000, and beat random assignment at equal budget. Khramtsova et al. [21] find label-free *collection-level* selection among dense retrievers on BEIR ineffective. Bruch et al. [6] show that convex combination (CC) beats reciprocal rank fusion (RRF) [22], that RRF is parameter-sensitive, and that tuning CC is sample-efficient. Dense retrievers benefit from BM25 interpolation [4], and zero-shot hybrids help out of domain [5]. In legal paragraph retrieval, Mori et al. [23] find that query-level traits modulate the BM25–dense gap. There, BM25 beats off-the-shelf dense models where verbatim quotes are *less* prevalent and on longer queries, while a dense model fine-tuned in-domain beats BM25 on most metrics.

III. EXPERIMENTAL SETUP

Collections. We use the test splits of eight BEIR datasets (Table I). The jargon rate is the share of corpus tokens with Zipf frequency below 3 in the `wordfreq` lists.

Systems.

- *Lexical*, implemented from scratch: Boolean AND/OR; TF-IDF with SMART `ltc` weighting [24]; and Lucene-style BM25 ($k_1 = 0.9$, $b = 0.4$, Porter stemming, a flat title+text field). BM25 is also run with RM3 feedback [25] and with WordNet expansion.
- *Dense*: all-MiniLM-L6-v2 ("MiniLM", 22M parameters) [26] and bge-small-en-v1.5 ("BGE-s", 33M) [27], with maximum length 256 and exact search. MiniLM rankings are available on CPU for five collections; for the other three, the dense results (and all BGE-s results) come from Colab as per-query scores only. Experiments that need both rankings therefore use those five collections.
- *Hybrid*: $\mathrm{H(D)} = 0.5\,\mathrm{minmax(B)} + 0.5\,\mathrm{minmax(D)}$ over the two top-1000 lists, never tuned; RRF ($k = 60$) is secondary.
- *Re-ranker*: the cross-encoder `cross-encoder/ms-marco-MiniLM-L-6-v2` ("CE") re-ranks a top-100.

Router features (no relevance labels). *Tier A* has 23 features: 15 pre-retrieval features (length, IDF, SCQ, VAR, avICTF, SCS, jargon and OOV statistics) and 8 BM25 post-retrieval predictors (WIG, NQC, SMV, Clarity and score gaps). *Tier B* adds 19 features: the same predictors on the dense list, plus BM25–dense list agreement (Jaccard, RBO, cross-ranks, pseudo-coverage). Tier B currently exists for MiniLM on five collections; the full list is in the code release. For explanation only, the *coverage* of a query is the mean share of its terms that occur in its relevant documents. Coverage needs labels, so it is never a router input.

Router. Per query, $\Delta_\mathrm{B} = \mathrm{nDCG_B} - \mathrm{nDCG_H}$ and Δ_D are regressed with Ridge or gradient-boosted trees; the model family is chosen by an inner LODO on the training collections. The pre-declared "selected" router picks $\arg\max\{0, \hat{\Delta}_\mathrm{B}, \hat{\Delta}_\mathrm{D}\}$, preferring the cheapest system (B < D < H) within 0.005 of the maximum. Secondary variants are Ridge or HGB alone, logistic classification, transductive percentile features, a LODO-tuned BM25, and plain argmax without the cost rule (added post hoc). The outer loop is LODO: train on seven collections and test on the eighth.

Statistics. The metric is per-query nDCG@10.

- The headline is the *macro* average over collections. It gives TREC-COVID (50 queries) and Touché (49) the same weight as Quora (10k), so we also report results without them. The pre-declared secondary aggregate is the micro (query-weighted) average.
- Per collection: paired randomization tests [28] (10^4 sign flips) with Holm's step-down correction within each declared family.
- Macro differences: a stratified sign-flip test, and a stratified percentile bootstrap 95% CI (10^4 resamples within collections).
- *Pooled* means a DerSimonian–Laird random-effects estimate [29]. With 5–8 collections these intervals are anti-conservative, so we treat the macro bootstrap as primary. The seed is 42.

Pre-declaration and leakage.

- H1 (BM25>TF-IDF>Boolean everywhere) and H2 (BM25 degrades with jargon) were fixed, with decision rules, in the Phase-1 analysis code; H2 is the motivating question of the project proposal.
- The router hypothesis (macro router − always-H > 0), the headline variant and the test families were written into an experiment log before any router result was computed.

TABLE II
nDCG@10. Ref: Pyserini BM25-flat [7]. H(D): CC hybrid with $\alpha = 0.5$. Best per row in bold.

Dataset	Ref	BM25	TF-IDF	MiniLM	BGE-s	H(Mi)	H(BGE)
NFCorpus	.322	.320	.316	.316	.343	.355	**.366**
SciFact	.679	.677	.646	.645	.701	**.731**	**.731**
FiQA	.236	.237	.175	.369	.391	**.394**	.391
SciDocs	.149	.150	.146	**.216**	.203	.204	.196
ArguAna	.397	.402	.534	.503	**.603**	.495	.548
TREC-COVID	.595	.595	.290	.472	.746	.694	**.794**
Quora	.789	.788	.785	.876	**.888**	.883	.887
Touché	.442	**.444**	.049	.169	.242	.357	.358
Macro	.451	.452	.368	.446	.515	.514	**.534**

- Four improvement tests (P1–P4) were also declared in advance. A claim counts as an improvement only if the Holm-adjusted one-sided $p < 0.05$ *and* the macro gain is at least $+0.005$.
- These logs are local files with no external timestamp; they are not registered with a third party.
- Everything else is exploratory.
- Leakage controls: feature files are machine-checked to contain no qrel-derived columns; scalers and models are fitted on training collections only; tuned BM25 parameters are chosen LODO; the cross-encoder teacher uses no target labels and is evaluated on held-out queries.

IV. Results

A. Baselines and the domain-level hypotheses

Table II gives nDCG@10. At three decimals, our BM25 is within 0.005 of the Pyserini flat reference [7] on all eight collections (mean $|\Delta| = 0.0018$). BGE-s is within 0.02 of its model card's scores [30].

H1 is not supported. BM25 beats TF-IDF with Holm significance on only 4/8 collections. TF-IDF wins on ArguAna (0.534 vs. 0.402), whose queries are whole arguments; BM25 tuned on that test set reaches 0.528. The pooled difference is $+0.064$ [0.025, 0.103], but -0.005 [-0.039, 0.029] without TREC-COVID and Touché. Boolean AND returns an empty set for 62–93% of queries on six collections.

H2 is not supported. Jargon rate and BM25 nDCG@10 correlate at $\rho = +0.33$ ($p = 0.42$); jargon and the MiniLM gain over BM25 correlate at $\rho = -0.60$ ($p = 0.12$). With $n = 8$, the minimum detectable $|\rho|$ ($\alpha = 0.05$, power 0.8) is about 0.85, so we report *no detectable* domain effect. FiQA has the largest MiniLM gain ($+0.132$) and one of the lowest jargon rates.

Table III reports the paired effects. Heterogeneity is extreme ($I^2 = 0.87$–0.99). The classical lexical fixes do not help on average. Leakage-free BM25 tuning gains $+0.005$ pooled (-0.002 macro) and costs Touché -0.112 [-0.147, -0.077]. MiniLM alone is not reliably better than BM25. The fixed hybrid is the only system robust in sign. H(BGE-s) beats BM25 with Holm significance on 7/8 collections and has higher recall@100 on all eight (7 Holm-significant); its only nDCG@10 loss is on Touché. CC slightly beats RRF (macro 0.534 vs. 0.528 for BGE-s), consistent with [6].

TABLE III
Paired differences in nDCG@10 over 8 collections. Macro: unweighted mean of per-collection differences. Pooled: DerSimonian–Laird with 95% CI. Sig. ±: collections with a Holm-significant gain/loss.

Comparison	Macro	Pooled [95% CI]	Signs	Sig. ±
BM25 − TF-IDF	$+0.084$	$+0.064$ [$+0.025$, $+0.103$]	7/1	4/1
Stemming (BM25)	$+0.000$	$+0.005$ [-0.017, $+0.026$]	6/2	2/1
RM3 − BM25	-0.004	-0.005 [-0.038, $+0.029$]	4/4	2/3
WordNet − BM25	-0.004	-0.003 [-0.007, $+0.001$]	4/4	0/1
BM25-LODO − BM25	-0.002	$+0.005$ [-0.002, $+0.013$]	6/2	4/1
MiniLM − BM25	-0.006	$+0.007$ [-0.033, $+0.047$]	4/4	4/2
BGE-s − BM25	$+0.063$	$+0.067$ [$+0.024$, $+0.110$]	7/1	5/1
H(MiniLM) − BM25	$+0.063$	$+0.066$ [$+0.041$, $+0.092$]	7/1	7/1
H(BGE-s) − BM25	$+0.082$	$+0.083$ [$+0.051$, $+0.115$]	7/1	7/1
H(BGE-s) − BGE-s	$+0.019$	$+0.011$ [-0.006, $+0.028$]	5/3	2/1

B. Query-level overlap: difficulty vs. lexical match

Within collections, coverage predicts BM25 nDCG@10 (partial $\rho = 0.37$–0.73 on 7/8; null on Touché), but coverage also measures how hard a query is for any retriever. We regress per collection, controlling for query length, IDF and the number of relevant documents, and pool. The coverage slope is 0.182 [0.144, 0.219] on BM25 nDCG@10, but also 0.120 on MiniLM and 0.119 on BGE-s nDCG@10. About two thirds of the association (ratio 0.66) is therefore shared difficulty. The lexical-specific remainder, $+0.059$ [0.046, 0.073] (MiniLM) and $+0.063$ [0.050, 0.076] (BGE-s), equals by OLS linearity the negative of the slope of the dense−BM25 gap on coverage. That gap slope is negative on 8/8 collections; it is the same evidence, not independent evidence. The deployable version fails: a label-free proxy (coverage of query terms in the dense top 10) correlates with coverage at $\rho = 0.39$–0.53, but its gap slope is -0.006 [-0.022, 0.011] on five collections.

QPP features behave the same way. The best pre-retrieval predictor (VAR_{max}) correlates with BM25, MiniLM and BGE-s nDCG@10 almost identically (Kendall $\tau = 0.130$, 0.119, 0.118). On the five collections with MiniLM lists, the post-retrieval picture is mixed. $s_1 - s_{10}$ is stronger on the MiniLM list than on the BM25 list (0.30 vs. 0.24), whereas WIG (0.14 vs. 0.17) and NQC (0.16 vs. 0.19) are weaker, partly in line with [19].

C. Query-time routing does not beat always-hybrid

Table IV gives the core result. With BGE-s, the per-query oracle lies $+0.064$ above always-H, 3.4 times the dataset-oracle headroom ($+0.019$). Yet the pre-declared router scores 0.525 against 0.534 ($\Delta = -0.009$ [-0.015, -0.003], one-sided $p = 0.99$). **The router hypothesis is not supported in any configuration.**

Touché drives the size of the loss. Without Touché the BGE-s router is at $+0.002$ [-0.003, $+0.006$]. Without Touché and TREC-COVID it is at $+0.004$ [-0.001, $+0.008$] ($p = 0.06$; Table VI). Routing is therefore not better than always-H,

TABLE IV

LODO routing, macro nDCG@10 over held-out collections. Router: the pre-declared selected variant. Δ: router − always-H, with stratified bootstrap 95% CI; p: one-sided sign-flip test of $\Delta > 0$. Holm±: collections where the router is significantly better/worse than always-H. Mix: % of queries sent to B/D/H. †BM25 tuned LODO (also inside H).

Features	D (#coll.)	always-B	always-D	always-H	ds. oracle	q. oracle	Router	Δ	95% CI	p	Holm±	Mix
Tier A	BGE-s (8)	0.452	0.515	**0.534**	0.553	0.598	0.525	−0.009	[−0.015, −0.003]	0.99	1/2	5/41/54
Tier A	MiniLM (8)	0.452	0.446	**0.514**	0.528	0.576	0.506	−0.008	[−0.012, −0.005]	1.00	0/5	5/6/89
Tier B	MiniLM (5)	0.357	0.410	**0.436**	0.440	0.494	0.426	−0.010	[−0.016, −0.005]	1.00	0/2	0/49/51
Tier A†	MiniLM (5)	0.363	0.410	**0.432**	0.437	0.491	0.420	−0.012	[−0.017, −0.007]	1.00	0/1	0/36/63

TABLE V

Pre-declared improvement tests (one-sided, Holm over P1–P4; pass: $p_{\text{Holm}} < 0.05$ and $\Delta \geq +0.005$). k: collections; CI: stratified bootstrap 95%.

Test (vs. baseline)	k	Δ	95% CI	p_{Holm}	Pass
P1 per-query fusion weight vs. $H_{0.5}$	5	−0.0069	[−0.011, −0.003]	1.00	no
P2 CE-teacher label-free weight vs. $H_{0.5}$	5	−0.0044	[−0.006, −0.003]	1.00	no
P3 CE re-rank of H(MiniLM) (sample)	5	−0.0397	[−0.057, −0.023]	1.00	no
P4 *in-domain* router vs. always-H (diagnostic)	8	+0.0169	[+0.010, +0.023]	$< 10^{-3}$	formally

but on the six larger collections a gain of up to $+0.008$ is not excluded. The pre-declared micro average is 0.765 vs. 0.761 ($+0.004$), carried by ArguAna's 1,406 queries; under the macro headline this is not a win.

Other variants. The best of 35 variant–configuration pairs, Ridge on Tier A with BGE-s, reaches $+0.0015$ ($p = 0.24$; 95% CI $[-0.002, +0.005]$), so at most the pre-declared minimum gain. Plain argmax and percentile features stay below always-H. Logistic classification mostly collapses to always-dense ($\geq$91% dense in five of seven configurations).

Per collection (Holm), the BGE-s router is better than always-H only on ArguAna ($+0.048$, routing 91% of queries to the dense model). It is worse on SciDocs (-0.004) and Touché (-0.082). On Touché it sends no query to BM25, plausibly because no training collection has BM25 far ahead of dense retrieval.

Diagnostics. Three checks locate the failure.

- Within held-out collections, Kendall τ between predicted and true Δ_{B} is only 0.03–0.11 (Tier A) and 0.12 (Tier B), and about 0 for Δ_{D}.
- The router beats random routing with the same per-collection mix by only -0.0001 to $+0.008$.
- An efficiency variant that sends confident queries to BM25 alone lies on the random-assignment curve (5.7–7.9% BM25-only at a loss of 0.003–0.008).

D. Pre-declared improvement attempts (P1–P4)

Table V reports the four pre-declared tests. P1–P3 need both rankings and therefore use the five collections with MiniLM rankings (none of them small).

P1, a per-query fusion weight. A Ridge model predicts α_q in $\alpha\,\text{minmax(B)} + (1-\alpha)\,\text{minmax(D)}$ from Tier A+B features (LODO). It loses -0.0069, with a Holm-significant loss of -0.026 on FiQA. A secondary HGB variant gains $+0.0027$ ($p = 0.11$); a single LODO-global weight gains $+0.0037$ (uncorrected $p = 0.03$; DL CI includes 0). Even the best weight per collection in hindsight gains only $+0.012$. The fusion weight barely matters, as Bruch et al. [6] found for tuning; we find the same for prediction.

P2, a label-free weight. The CE scores 150 unlabelled target queries, and its scores serve as pseudo-gains for choosing $\hat{\alpha}$. The selector picks 0.5 on 3/5 collections. On SciDocs it goes the wrong way (0.7 vs. an oracle of 0.2): there, the pseudo-metric is anti-correlated with true nDCG@10 across α ($r = -0.49$).

P3, a stronger model. Re-ranking the hybrid's top 100 with the 22M-parameter CE *hurts* (-0.040; ArguAna -0.120). Re-ranking BM25's top 100 helps on 8 collections: $+0.025$ $[+0.009, +0.041]$, $p = 0.004$, Holm $+4/-1$; TREC-COVID $+0.143$, Touché -0.164 $[-0.230, -0.100]$. Its direction matches BEIR's BM25+CE results on seven collections and is null on ArguAna in both [3]. CE(BM25) still trails the untuned H(BGE-s) by -0.059 (5/8 Holm-significant). The CE runs use at most 150 queries per collection (all 50/49 for TREC-COVID/Touché).

P4, the in-domain ceiling. If query-level signal exists but does not transfer, a router trained in-domain (5-fold CV inside each collection, same features and model selection) should beat always-H. It does: $+0.0169$. But it ties with the in-domain *best fixed system* (-0.0005 $[-0.003, +0.002]$, $p = 0.66$; Fig. 1). Essentially all of the gain is the collection prior: dense retrieval for ArguAna and SciDocs, BM25 for Touché. The same holds for MiniLM with Tier A (-0.0036 vs. fixed) and Tier B (-0.0016). The router beats the fixed system only on Quora, which has about 8k training queries per fold: $+0.0046$ [0.0026, 0.0066], Holm $p = 0.0008$. P4 uses target labels, so it is a diagnostic, not a deployable improvement.

E. How much of the oracle headroom is real?

Part of the per-query headroom is measurement noise. Consider BM25 with and without stemming, whose macro nDCG@10 differs by only 0.0004 (although by up to 0.09 on single collections). An oracle choosing between them per query already gains $+0.043$ over BM25. For BM25 vs. BM25+WordNet, the gain is $+0.013$. Against these refer-

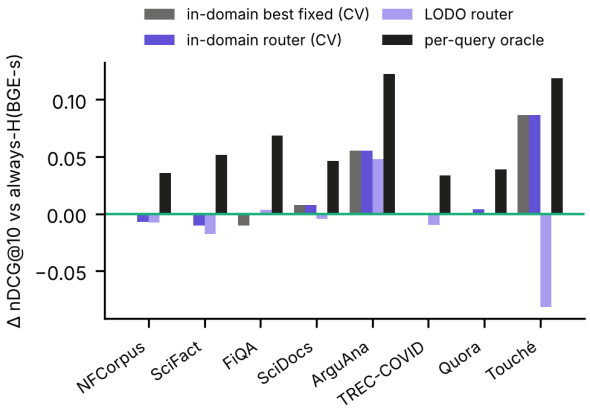


Fig. 1. Gain over always-H(BGE-s). The in-domain CV router (purple) matches the in-domain best fixed system (grey); the LODO router (light purple) loses on Touché; the per-query oracle (black) is out of reach.

ence points, the $+0.064$ three-way headroom is modest. The collection-level headroom, by contrast, is realisable: choosing the best fixed system on a random half of each collection and evaluating it on the other half (1,000 splits) gains $+0.0182$ for BGE-s, close to the in-sample $+0.0188$; the query oracle has no such out-of-sample check.

The headroom is also concentrated. For BGE-s, the queries with an oracle gain above 0.2 are 2–27% of all queries but carry 23–87% of the headroom. On ArguAna, SciFact and Quora the median query has one relevant document, and queries with at most two carry 83–100% of the headroom. There, a per-query "preference" amounts to whether one document crosses rank 10.

F. Collection-level selection with few labels (exploratory)

In a post-hoc analysis, we choose one system (B, D or H) per collection from k random labelled queries and evaluate it on the rest (Fig. 2). For BGE-s this gains $+0.004$, $+0.010$, $+0.014$ and $+0.016$ at $k = 5, 10, 20, 30$; for MiniLM, -0.004, $+0.005$, $+0.009$ and $+0.010$. These gains depend on Touché (Table VI). Without it, BGE-s gains -0.006, 0.000, $+0.004$ and $+0.006$, and MiniLM is negative for every $k \leq 30$. Touché, where BM25 is selected, is also the collection whose pool favours BM25 (Sec. VI), so the gain may partly reflect judgement bias.

V. Discussion

Why routing fails. Three explanations fit the evidence.

- *Difficulty, not preference.* The features predict how hard a query is for every retriever, whereas the router needs the *difference* between retrievers ($\tau \leq 0.12$). This is consistent with Faggioli et al. [19].
- *Noisy, concentrated labels.* The per-query headroom is comparable to what choosing between near-identical systems produces, and it sits in rare swings on queries with one or two relevant documents.

TABLE VI
Small-collection sensitivity: macro Δ nDCG@10 on all 8 collections, without Touché, and without Touché and TREC-COVID. Stratified bootstrap CIs for rows 1–4 are in numbers_trace.md (row 1 also in the text). Rows 5–7 are exploratory.

Contrast	All 8	−Touché	−Touché −COVID
Router − H (BGE-s)	−0.009	+0.002	+0.004
Router − H (MiniLM)	−0.008	−0.010	−0.007
In-domain router − H (BGE-s)	+0.017	+0.007	+0.008
CE(BM25) − BM25 (sample)	+0.025	+0.052	+0.037
Split-half choice (BGE-s)	+0.018	+0.008	+0.010
k=20 choice (BGE-s)	+0.014	+0.004	+0.005
k=20 choice (MiniLM)	+0.008	−0.003	−0.003

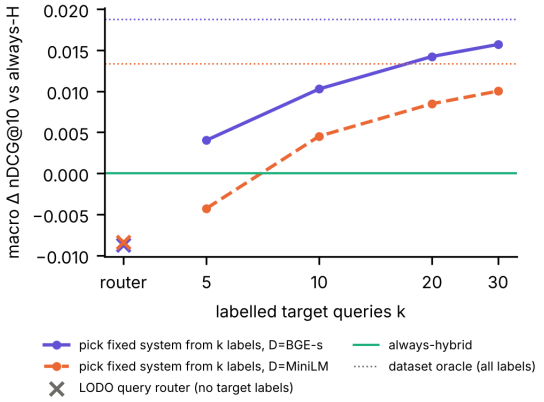


Fig. 2. Exploratory: macro gain over always-H when one fixed system is chosen per collection from k labelled queries (mean of 1,000 draws). Crosses: LODO query routers. Most of the gain comes from Touché (Table VI).

- *Prior shift.* What transfers is a collection-level prior, and LODO cannot learn a prior that no training collection shows (Touché). This prior may itself be partly a pooling artefact: dense judged@k on Touché is pending.

The in-domain ceiling shows that the failure is not only one of transfer. Even in-domain, these features learn only the prior, except where many labels exist (Quora).

Practical guideline. Deploy an untuned convex hybrid of BM25 and a dense model. A handful of labelled target queries can detect the rare collection where one system dominates (here, BM25 on Touché); otherwise, expect no gain. Do not transfer tuned BM25 parameters across domains: tuning on other domains costs Touché -0.112. Do not assume a small cross-encoder improves a hybrid first stage.

Positioning. Our setting differs from Arabzadeh et al. [20] in data (zero-shot across BEIR vs. in-domain MS MARCO), router (hand-crafted QPP features vs. a fine-tuned BERT), metric and hybrid, so we do not contradict them. Our Quora fold, the one with abundant in-domain labels, shows a small learnable query-level signal. We complement Khramtsova et al. [21] at the query level: here too, selection is ineffective. Our hybrid's exceptions (Touché, ArguAna, Quora and SciDocs) largely coincide with those Kamalloo et al. list for their min-max hybrid [7]. Our result concerns cheap QPP features;

routers built on learned query encoders or large language models are not tested here. Mori et al. [23] report query-level modulators of the BM25–dense gap in a legal setting. We find a qrel-based modulator in all eight domains, but its label-free proxy fails. Their constructs differ from ours, so the two results neither confirm nor contradict each other.

VI. Threats to Validity

- **Power and weighting.** Eight collections can detect only $|\rho| \gtrsim 0.85$. The macro headline gives two 50-query collections a quarter of the weight, so we report sensitivities (Table VI).
- **Judgement pools.** On Touché, only 47% of BM25's top 10 is judged, and BM25's nDCG@10 rises from 0.444 to 0.824 on condensed lists. On NFCorpus, BM25's judged@100 is 0.186 vs. 0.080 for MiniLM. All Touché statements carry this caveat [8].
- **Model strength and router capacity.** Our dense models are small, and the CE results come from one small reranker on a query sample. Our routers are linear or tree models on hand-crafted features. On TREC-COVID and Touché, the in-domain tree router is constant by construction (40-query folds, leaf size 50). The negative result is bounded to these choices.
- **Configuration and statistics.** Flat and multifield BM25 differ by up to 0.077 on our collections. The router macro tests are not corrected across 35 variants; this does not matter for a negative claim.
- **Post-hoc analyses.** The argmax router variant, few-shot selection, sensitivities, noise reference points and bootstrap CIs were added after the main results. A single-annotator failure taxonomy from an earlier phase is omitted.

VII. Conclusion

Across eight BEIR domains, the choice among BM25, dense and hybrid retrieval is predictable at the level of the collection but not at the level of the query. The per-query oracle headroom is 3.4 times the collection-level headroom, but much of it is of the size that per-query noise alone produces. No router built on standard QPP and list-agreement signals realises its query-level part, whether it routes discretely, predicts a fusion weight, learns from a cross-encoder teacher or is trained in-domain. These signals measure query difficulty, which all retrievers share. A fixed hybrid is the safe default; a handful of labelled target queries can detect the rare collection where one system dominates, which query routing cannot.

Reproducibility

The code covers indexing, retrieval, metrics, features, routers, statistics and figures. It runs on 4 CPU cores; GPU encodings come from the provided Colab notebooks. The release also includes the per-query result files, the experiment logs with their pre-declarations, and scripts for every number (seed 42; 10^4 permutations and resamples). The code, notebooks and per-query result files are available from the author on request.

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- designed and ran the experiments and statistical analyses, and wrote the analysis code and the pre-declarations in the experiment logs;
- searched and verified the literature;
- drafted and revised this manuscript.

The author directed the work, ran the GPU notebooks in Google Colab, and takes full responsibility for the content.

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